

Worksheet 1.1–1.2

Name: _____

1. Function operations and domain. Let

$$f(x) = \sqrt{x+2}, \quad g(x) = \frac{x-1}{x+3}.$$

Find $\left(\frac{f}{g}\right)(x)$ and state its domain in interval notation. Your domain must account for the domains of both original functions *and* the fact that the divisor cannot equal zero.

2. Information from a formula. Let

$$h(x) = \frac{x^2 - x - 6}{x + 1}.$$

Find the domain, all x - and y -intercepts, and $h(-2)$. Then state whether the point $(3, 0)$ lies on the graph and justify your answer.